S T R U C T U R E D P E E R R E V I E W

Peer Review Report

Manuscript: “Digital Twins for Cyber-Resilient Power Grids: A Systematic Review and a Research Agenda for the AI-Powered Era”

> Manuscript file: manuscript.pdf (29 pages, 26 references)
>
> Review basis: Full-text read-through with arithmetic and methodology verification
>
> Overall recommendation: Major Revision
>
> Review date: October 16, 2026

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1. Executive Summary

The manuscript under review reports a PRISMA-2020-style systematic literature review of digital twins for power-grid cybersecurity, built from 1,287 unique OpenAlex records (a pool of approximately 2,000 after four adversarial-AI probe queries) and distilled into a retained corpus of 100 studies published between 2020 and 2026. The review is explicitly theory-testing: it operationalises the hypothesis that grid digital-twin security research adopts artificial intelligence (AI) as a defensive tool far faster than it models AI-enabled threats, or the twin itself, as an attack target. The headline measurements are that AI is used defensively in 70 of 83 full-text studies; that 35 of 83 (42.2%) engage at least one AI-enabled threat at any level while only 11 (13.3%) do so substantively; that the twin is an explicit attack target in exactly one study; and that threat coverage concentrates at the operation lifecycle phase (66 of 83), with one study coded at design and none at commissioning. A five-item research agenda closes the paper.

My overall assessment is positive on intent and transparency, but the manuscript is not yet ready for acceptance. Its most distinctive virtue is candour: inter-coder reliability is reported field by field including the fields where agreement is weak, post-hoc analytical choices are flagged as such, and the central absence claim is audited with a two-lexicon false-negative exercise that includes a positive control. This level of self-scrutiny is rare in engineering systematic reviews and deserves recognition. Against this, four substantive problems remain: **the corpus is assembled from a single database and cut to an arbitrary top-100 by a non-probabilistic ranking that heavily favours recency; the reliability of the fields the headline depends on is moderate at best (Cohen's $\kappa = 0.41$ for overall AI-threat engagement and $\kappa = 0.14$ for substantive engagement on a 20-study sample); several of the sharpest claims are partly predetermined by conservative codebook definitions; and the referencing, abstract and PRISMA flow details are not submission-ready.**

I recommend Major Revision. The direction of the central finding — a defensive bias in the grid digital-twin security literature — is credible, survives the paper's own audits and sensitivity checks, and is worth publishing once the magnitudes are framed honestly, the reliability of the headline fields is strengthened or explicitly bounded, and the reporting apparatus is completed. Section 7 provides a prioritised, itemised revision checklist, and Table 1 summarises my assessment by dimension.

Table 1: Overall assessment scorecard

Dimension	Assessment	Key evidence (manuscript location)
Research question and motivation	Strong	Falsifiable asymmetry hypothesis; RQ1–RQ5 map onto results; timely intersection of twins, AI and grid security (§1, §4.1)
Methodological transparency	Exemplary	Field-by-field κ incl. unfavourable values; post-hoc readings flagged; coder-1 precedence declared (§4.8–4.9, §5.6, §8)
Corpus construction	Weak–Moderate	Single database; non-probabilistic top-100 cut; 76% of corpus from 2025–26; no search date or registered protocol (§4.2–4.4, §8.4–8.5)
Reliability of headline measures	Moderate	ai_threat_overall κ = 0.41; substantive κ = 0.14; n = 20 sample; coder 1 coded 80% of corpus (§4.8, Table 5)
Statistical treatment	Moderate	Wilson intervals on a non-random sample; no test for the review-share comparison; post-hoc κ collapses (§4.3, §5.4–5.6)
Positioning against prior reviews	Strong	Additive vs Habeeb et al.; vocabulary-to-vector mapping; coverage-vs-treatment caveat (§3, Table 1)
Primary-study quality appraisal	Weak	No formal risk-of-bias instrument; low-barrier venues in corpus; 18 of 54 venue labels corrected against Crossref (§4.11, §8.7)
Reproducibility	Strong	96 of 97 DOIs verified; codebook, coding sheets and scripts released; missing-not-zero policy (§4.5, §4.11)
Presentation and referencing	Weak	Venue placeholders in most references; duplicated sentence; broken tokens; internal note left in Highlights (§5.5, Highlights)

2. Manuscript Summary

Methodologically, the review proceeds in four stages. Identification combined twelve OpenAlex core queries pairing a digital-twin term, a grid-domain term and a security term, yielding 1,287 unique records after de-duplication; four additional adversarial-AI probes (adversarial machine learning, poisoning, evasion, and large-language-model terms) widened the candidate pool to approximately 2,000 records. Title-and-abstract screening required at least one term from each of three classes (digital twin; grid or energy; cyber or

security), leaving 492 eligible records, which were narrowed to 100 by a ranking combining OpenAlex relevance score, citation count and recency, with the final cut at rank 100. Of the retained 100, 83 were coded at full text across 46 fields using computer-assisted but human-decided extraction; 17 were available only as abstracts and are carried with abstract-level fields treated as missing rather than zero.

The results are presented figure-led against five research questions. On positioning (RQ1), the adjudicated stance distribution is defensive 42, review 36, bridge 10, enabling 10 and adversarial 2, and the twin_role field codes the twin as a defensive tool in 75 of 83 full-text studies, as an attack target in one, as both in zero, and as unclear in seven. On threat vocabulary and lifecycle (RQ2), denial of service (49 records), false data injection (46) and malware (34) lead a closed vocabulary; a mean of 3.77 distinct threats is named per study; explicit threat models appear in 19 of 83 studies and adaptive adversaries in 10; and 66 of 83 studies sit at the operation phase. On AI-threat engagement (RQ3), 35 of 83 engage any AI-enabled threat and 11 substantively, led by adversarial machine learning (24/8 at any/substantive level) and poisoning (19/7). On evidence quality (RQ4), machine learning is used in 70 of 83 studies, predominantly for detection; evaluation is simulation-heavy (33) with only two hardware-in-the-loop studies; maturity is prototype-heavy (46) with three validated; and code is released in 3 of 83 studies and data in 16. A two-lexicon false-negative audit, a 20% blind reliability sample and a DOI and Crossref verification pass support the numbers.

The paper claims seven contributions, from the corpus and extraction method through the headline asymmetry, the lifecycle concentration and a vocabulary mapping onto the five attack vectors of the closest prior review (Habeeb et al., 2026), to a five-item research agenda (P1–P5) whose items are each tied to a quantified gap and a validation criterion. The authors are careful to scope their claims: the review is positioned as a grid-side lower bound on the attack-surface axis named by Habeeb et al.; the vocabulary mapping is described as coverage rather than evidence of empirical treatment; and the agenda is framed as measuring research attention, not deployment security. These scoping statements are a genuine strength; my concern, developed in Section 5, is that some of the abstract-level and highlight-level phrasing does not carry the same qualifiers as the body text.

3. Review Scope and Method

This review is based on a complete reading of the 29-page manuscript, including all tables (Tables 1–6), figures (Figures 1–5), the highlights, the research agenda and the reference list. I verified the internal arithmetic of the principal counts: the stance distribution sums to 100; the lifecycle, evaluation-method, maturity, ML-role and grid-domain distributions each sum to 83; the Figure 4 threat-by-lifecycle row sums reproduce the Table 3 threat

counts (for example, false data injection 37 + 7 + 2 = 46 and denial of service 35 + 11 + 3 = 49); the screening arithmetic reconciles (1,287 − 795 = 492); and the reported percentages (35/83 = 42.2%, 11/83 = 13.3%, 48/83 = 57.8%, 35/70 = 50%) are all correct. I assessed conformance against the PRISMA 2020 reporting standard and standard practice for inter-coder reliability in systematic reviews, and I examined the statistical presentation of proportions, intervals and κ statistics.

Two limitations of this review should be noted. First, the submission does not include the released reproducibility package, so I could not audit the coding sheets, the reliability script or the raw pre-adjudication codes; my assessment of those artefacts relies on the manuscript's descriptions of them. Second, I did not re-run the searches or attempt an independent retrieval of the attack-side literature, so statements about search sensitivity rest on the paper's own probe design and audit rather than on independent verification. Neither limitation affects the copy-editing findings, the reliability analysis or the design-level criticisms, all of which rest on the manuscript itself.

4. Key Strengths

The manuscript has seven substantial strengths, four of which — the falsifiable design, the transparent reliability reporting, the false-negative audit and the reproducibility apparatus — are, in my experience, well above the norm for systematic reviews in this field. I detail each below with references to the manuscript.

4.1 A falsifiable question and a genuine theory-testing design

Most systematic reviews in the digital-twin domain are descriptive maps: they aggregate papers, name themes and stop. This manuscript instead commits to a specific, falsifiable hypothesis — that the literature adopts AI as a defensive tool far faster than it models AI-enabled threats or the twin as a target — and operationalises it with a closed threat vocabulary, a 0–2 ordinal engagement scale applied to full text (§4.7) and pre-stated research questions RQ1–RQ5 that map one-to-one onto the results sections. The design choice that the retained 100 studies "are the test" (§4.1) makes the review answerable: a reader can see exactly what would confirm or disconfirm the asymmetry. The screening-seed versus read-coding comparison (Figure 3b), showing that keyword-level screening over-called an adversarial stance (11 seeds against 2 after reading), is a particularly instructive demonstration that the measurement genuinely required reading rather than string matching.

4.2 Reliability is reported field by field, including the unfavourable values

The paper reports Cohen's κ for 18 interpretable fields, percentage agreement for near-constant fields and Jaccard or set metrics for multi-label fields (Table 5), and it does so without suppressing the weak ones: overall AI-threat engagement $\kappa = 0.41$, model evasion $\kappa = 0.39$, lifecycle stage $\kappa = 0.32$ and attack-surface-named $\kappa = 0.14$ all appear in the abstract, the results and the threats-to-validity section. The post-hoc collapse of the 3×3 agreement table is explicitly "flagged as such" (§5.6); the systematic default-category difference between coders is attributed to a codebook problem rather than a coder problem; and the substantive-engagement count is labelled "fragile" in the authors' own highlights. The abstract's phrase "a result we report rather than hide" captures the spirit. This is the single most impressive feature of the manuscript, and it materially helps readers calibrate trust in every downstream number.

4.3 The retrieval design anticipates the attack-side blind spot

The authors recognised that a grid-security query vocabulary retrieves defenders, not attackers, and added four adversarial-AI probe queries for precisely that reason (§4.2). The stated logic — that the asymmetry result should be robust to retrieval bias — is exactly the right instinct, and §8.5 correctly concludes that even with the probes the engagement remains shallow, so the asymmetry is "not merely a retrieval artefact". Few reviews in this area consider retrieval-side bias at all; embedding the mitigation in the search design, rather than leaving it as a discussion-section apology, is good practice and strengthens the paper against the most obvious attack on its headline.

4.4 The headline absence is audited rather than asserted

Because the paper's central claim is an absence, the authors audit it (§5.7). A strict AI-threat lexicon run over the 48 records coded as non-engaging flagged none — but the authors also report the lexicon's positive control, recovering only 7 of 11 known positives, and therefore do not over-claim confirmatory force. A broader eleven-family net then fired on 38 of the 48 negatives with 23 strong hits, all of which the authors inspect and clear with documented, plausible false positives: the semiconductor GaN matching "GAN" in two records, the author surname "Gan", Xenon-135 reactor physics matching "poisoning", ARP poisoning, and the SolarWinds incident discussed as a data-poisoning example. The closing statement — that the audit "bounds but does not eliminate" the false-negative channel — is precisely the right epistemic posture, and the GaN/GAN confusions double as a persuasive argument for human coding over keyword coding.

4.5 A real reproducibility apparatus

Every DOI was verified against an authoritative registry (96 of 97 resolve with a title match); corpus metadata was cross-checked against Crossref with disagreements resolved in favour of the registry; the codebook's five revisions are logged and dated; the raw pre-adjudication codes are released frozen alongside the adjudicated matrix; and the extraction treats abstract-only fields as missing rather than zero (§4.5, §4.11, §8.6). The 17% unobtainable full-text rate is reported as a finding in its own right, with DOIs listed so readers can check what is missing. This apparatus makes the review unusually checkable, and the disclosure of failures — one unregistered DOI, three records without DOI, 18 of 54 venue-label disagreements — increases rather than decreases confidence in the parts that do check out.

4.6 Positioning against prior work is disciplined and additive

The relationship to the closest prior review — Habeeb et al.'s 2026 PRISMA review of dual-role digital twins — is handled with unusual care. The authors explicitly decline credit for discovering the dual-role tension, position their work as "a lower bound on its grid manifestation" (§1.1), and devote Table 1 to mapping their independently designed threat vocabulary onto Habeeb's five attack vectors, with the explicit caveat that vocabulary coverage "is not evidence that the vectors are dealt with empirically" (§1.2). Section 6.5 adds two structural caveats — the prior review's query lacks attack-side operators and its corpus is cross-domain — and warns against generalising in either direction. This is a model of non-overclaiming related-work positioning, and it makes the contribution boundary easy for an editor to verify.

4.7 The research agenda is testable, and the reported numbers cohere

Each agenda item names the quantified gap that justifies it and the evidence that would validate it (Table 6): P1 (twin as target) is justified by twin_role target = 1 and validated by a reproducible twin-directed attack with released code and a stated adaptive adversary; P5 (release code and data) is justified by the 3/83 and 16/83 release rates and validated by unaided third-party reruns. The P5 caveat — that the review measures research attention, not deployment security — is exactly the qualifier that prevents misuse of the headline. Finally, I verified the internal arithmetic of the principal counts (Section 3 above): distributions sum correctly across stance (100) and across lifecycle, evaluation method, maturity, ML role and grid domain (83 each), and the threat-by-lifecycle heatmap reproduces the threat-vocabulary table exactly. Whatever else one says about the design choices, the numbers that are reported do cohere.

5. Key Weaknesses (Major Concerns)

I group the major concerns into eight items. W1–W3 go to the validity of the headline measurements; W4–W6 go to scope and reporting completeness; W7–W8 go to appraisal and presentation. None is fatal, but each requires substantive revision rather than wording alone.

5.1 W1 — Corpus construction and sampling frame undermine the prevalence claims

The corpus is drawn from a single database (OpenAlex), with no stated search date and no registered protocol — limitations the authors acknowledge (§8.4) but which nonetheless place real distance between the manuscript and the PRISMA 2020 standard it invokes. More consequentially, the eligible set of 492 was cut to exactly 100 studies by a non-probabilistic ranking of OpenAlex relevance, citation count and recency (§4.3). Three problems follow. First, the round-number threshold is arbitrary, and no sensitivity analysis varies the cut (for example, top 150 or top 200) to show whether the headline proportions are stable. Second, because 76% of the corpus was published in 2025–2026 (45 records in 2026 alone), citation count is nearly uninformative for most of the corpus, so the ranking is effectively relevance-plus-recency; the authors concede that foundational 2020–2023 attack-side work “may be under-weighted” (§8.5) — precisely the literature whose absence the paper measures. Third, Wilson 95% intervals (32–53%, 8–22%, 47–68%) are attached to proportions from a non-random, purposive sample. A binomial confidence interval models random sampling from a population; the retained 100 are not a random sample of anything, and the paper itself instructs that the figures “should not be read as an estimate of the wider literature” (§4.3). Presenting intervals in the abstract, results and conclusion anyway invites exactly that reading, and the intervals convey a precision the design cannot support. My recommendation: keep the percentages with an explicit “of the retained corpus” qualifier at every headline use, drop the intervals or re-derive them under a stated superpopulation assumption, and add a cutoff sensitivity analysis.

5.2 W2 — Reliability of the headline fields is moderate at best, and coder-1 dominance is structural

The headline field ai_threat_overall is coded at $\kappa = 0.41$ on a 20-study blind sample, and the “substantive” threshold — the basis of the 11/83 count — collapses to $\kappa = 0.14$, with only one of the six substantively coded studies agreed by both coders (the single (2,2) cell). With $n = 20$, the large-sample standard error of κ is of the order of 0.2–0.3 by the authors’ own calculation (§4.8), so Table 5 cannot stably rank field difficulty and the point estimates on the headline fields carry wide uncertainty. The paper is candid about all of

this — and still leads with the 11/83 figure in the abstract, the contribution list and the highlights, where “fragile” is the only hint. A second, structural issue compounds this: coder 1 coded 80% of the corpus, adjudication defaults to coder 1 where both values are defensible (§4.9), and coder 1 exhibits a systematic default-category preference (high, yes, operation, single; §5.6). The lifecycle field — κ = 0.32, the lowest interpretable κ in the table — is exactly where this default bias bites, because “operation” is coder 1’s default. The 66/83 operation concentration and the design and commissioning void that anchors agenda item P2 may therefore be partly a coding artefact, a possibility the authors concede in §6.3 but do not carry into the agenda’s framing. My recommendation: enlarge the blind dual-coded sample to at least 40% for the headline fields (ai_threat_overall, twin_role, lifecycle_stage), report κ with bootstrap confidence intervals, assign headline-field adjudication to the second coder or a third party, and restate P2 with the κ = 0.32 caveat attached.

5.3 W3 — Codebook strictness partially manufactures the sharpest claims

Several of the paper’s most quotable results are, by the authors’ own description, definitional outcomes. The twin_role rule requires positive evidence, and “both = 0 is therefore definitional: a paper cannot score both without stating the tension” (§4.6); “target” requires the twin to be the adversary’s objective rather than an asset under protection; and attack_surface_named requires a named interface or protocol, excluding passing mentions (codebook revision R3). Given these rules, the findings that the twin is an attack target in “exactly one study”, that no paper articulates the dual role, and that attack-surface naming is rare are partly baked in before any reading occurs. The authors do frame the review as a lower bound (§1.1), but the abstract and highlights present “exactly one study” and “both = 0” as empirical discoveries without the definitional qualifier, and a casual reader will not reconstruct it from the body text. My recommendation: either soften the phrasing to “at most one study under a strict codebook definition”, or — better — run a small sensitivity coding of the 83 full-text records under a deliberately looser twin-target definition and report how the counts move. This would convert a weakness into an interesting measurement of definitional sensitivity.

5.4 W4 — Scope paradox: the design measures attack-side engagement inside a defensively selected sub-literature

The eligibility rule requires papers to name both a digital twin and a security concern, and the manuscript acknowledges more than once that substantial adversarial-ML work on grid state estimation lies outside the corpus (§1, §8.7), as does work that functionally twins a grid without using the phrase (§2.3). This is defensible and disclosed — the abstract states

the restriction — but it creates a scope paradox that the headline framing does not escape: the review systematically excludes much of the very literature most likely to engage AI-enabled threats, and then measures that engagement as low. The title ("Digital Twins for Cyber-Resilient Power Grids"), the highlights and several agenda items generalise past the constructed corpus, and the one-study twin-as-target count leans on a corpus that was pre-filtered toward papers treating twins as instruments of defence. My recommendation is not to abandon the scoping rule but to make the restriction sentence do more work: attach it to every headline number, including the highlights, and frame the claim explicitly as being about the twin-naming sub-literature rather than grid cybersecurity at large.

5.5 W5 — Denominator churn and mixed coding depths impair comparability

The paper moves among at least five denominators — 100 (stance, all records), 83 (full-text fields), 70 (ML users), 49 (non-review) and 62 (journal articles) — without a consolidated guide to which n supports which claim. Figure 3a mixes full-text and abstract-only records in a single stance chart even though stance is coded at only $\kappa = 0.46$ with 60% raw agreement, and abstract-level stance coding is inherently weaker than full-text coding. Relatedly, Table 3 reports poisoning at 22 records while the AI-threat field ai_poisoning reports 19 at any level — a three-record discrepancy between two near-identical constructs that is nowhere explained — and the "wider relevant set" of 469 probe records used for the 5.1% abstract-level figure is never defined. The 5.1% versus 42.2% juxtaposition is disclaimed twice (§5.4, §8.2), yet pairing the two numbers in the results section still invites the "order-of-magnitude screening loss" quotation the authors themselves prohibit (§6.4). My recommendation: add a denominators table at the head of Section 5 (which n, which fields, why), reconcile or footnote the poisoning 22-versus-19 difference, define the 469-record set, and consider dropping the 24/469 comparison entirely, since the point it illustrates is already made qualitatively.

5.6 W6 — PRISMA flow and record-keeping gaps

The flow reporting has three specific gaps beyond the acknowledged absence of a protocol registration and a search date. First, Figure 1 states that screening excluded 795 records for having "no digital-twin term or no grid domain" — two exclusion reasons — while §4.4 specifies three required term classes (digital twin; grid or energy; cyber or security); the figure and the text disagree. Second, the probe yield is not reconciled anywhere: the core pool is 1,287 records, the probes raise the pool to approximately 2,000, yet the screening arithmetic (795 excluded, 492 eligible) reconciles only against 1,287, leaving the disposition of roughly 700 probe-added records unreported; the authors acknowledge that probe-only membership in the retained 100 "is not summarised" and call it "a gap in the

record” (§4.3, §8.5). Third, the “wider relevant set returned by the probes” of 469 records appears once, undefined, in §5.4. For a paper whose brand is candour, these are self-inflicted wounds: a revised PRISMA diagram with explicit probe strata (core versus probe-only, per-stage counts), a stated search execution date and a one-line query log per probe would close all three gaps.

5.7 W7 — No formal appraisal of primary-study quality or risk of bias

The corpus deliberately spans venues “of varying strictness (including IRJMETS and GSC Online Press)” (§8.7), yet no formal risk-of-bias or quality-appraisal instrument is applied, so the evidence-quality results — three validated studies, code released in three — blend intrinsic study quality with venue accessibility. The venue-quality sensitivity (journal-only subset) partially addresses this, but journal status is a weak proxy for methodological quality, and the corpus metadata itself needed correction: 18 of 54 venue labels disagreed with Crossref and were resolved in the registry’s favour (§4.11), a one-third disagreement rate that signals OpenAlex metadata noise in exactly the fields feeding the venue-type analysis. The anonymous journal article in the corpus (reference [19]) underscores the point. My recommendation: apply a lightweight appraisal instrument (for example, an adaptation of MMAT suited to the mixture of quantitative, qualitative and review studies) and report prevalence with and without studies failing minimum quality criteria; if that is out of scope, elevate the non-appraisal disclosure from §8.7 into the abstract-level limitations so no reader misses it.

5.8 W8 — The manuscript is not yet submission-ready at the presentation level

A dedicated copy-editing pass is required before this manuscript can be evaluated on its merits. The most substantive items are bibliographic: most reference entries carry a literal “journal (2026)” placeholder in place of the venue name, which is disqualifying in its current form for a review whose contribution is bibliographic discipline; reference [7] — the anchor of the entire positioning — lists “W. Habeeb, Almaliki” with the third named author (Weli) missing; and reference [19] is credited to “Anonymous”. In the body, §5.5 contains a fully duplicated sentence (“The evidence base is, in short, prototyping-heavy and release-light.”) together with a missing space (“corpus.evidence”); the phrase “engage-ment” appears with a spurious hyphen-space; and “twins targeted by one” (§5.5) is ungrammatical. The abstract runs to well over 400 words — far beyond the 250–300 words typical of journal limits — and contains a misplaced parenthetical (“subject to n = 20”) plus a confusing κ-collapse sentence in which agreement both rises (0.41 to 0.60) and falls (0.41 to 0.14) under the single word “collapsing”. The Highlights list contains the parenthetical “(fragile, original)”, which reads as an internal note left in place, and the

CRediT statement ends “(To be finalised.)”. Finally, the Data availability statement promises a reproducibility package without a URL or DOI, which for this particular manuscript is the single most important missing link. None of these defects is individually serious; collectively they signal a draft that has not been proofread end-to-end, which matters for a paper whose persuasive force depends on procedural care.

6. Minor Issues and Copy-Editing Corrections

Table 2 lists the specific, localised defects I identified, with suggested actions. Section and page references use the manuscript’s own numbering. Most items are mechanical; a few (rows 8, 10, 11 and 13) involve substantive reporting choices that duplicate points raised in Section 5, and are repeated here so that a copy-editor can close them in a single pass.

Table 2: Minor issues and suggested actions

#	Location	Issue	Suggested action
1	Abstract (p. 1)	Misplaced parenthetical “(subject to n = 20)” interrupts the reliability sentence	Move the sample-size qualifier next to “20% blind sample”; shorten the abstract to about 250 words
2	Abstract (p. 1)	“Agreement ranged from moderate to substantial” contradicts the κ = 0.14 quoted three lines later	Reword to “from slight to substantial, depending on the field”
3	Abstract (p. 1)	“Collapsing to κ = 0.60 … and κ = 0.14”: one value rises and one falls; the direction is unclear	State the two collapses separately: κ = 0.60 for any engagement; κ = 0.14 for substantive engagement
4	Highlights (p. 24)	“(fragile, original)” is an internal note, not reader-facing text	Delete the parenthetical from the Highlights
5	§5.5 (p. 16)	Sentence “The evidence base is, in short, prototyping-heavy and release-light.” duplicated; missing space in “corpus.evidence”	Delete the second occurrence; fix the spacing
6	§5.4 (p. 16)	Broken token “engage- ment”	Restore “engagement”
7	§5.5 (p. 16)	“Twin targeted by one” is ungrammatical	Reword, e.g. “the twin targeted in one”
8	§5.5 and §5.8 vs Table 4	Bracketed “[16, 17]” is ambiguous between bibliography entries and corpus record IDs	Adopt an unambiguous corpus-ID notation (e.g. “C16, C17”) wherever corpus IDs are cited

#	Location	Issue	Suggested action
9	References (pp. 25–27)	Venue rendered as a literal “journal (2026)” placeholder in most entries; [7] missing an author (Weli); [19] authored “Anonymous”	Complete all venue names and author lists from the DOIs
10	Figure 1 (p. 8) vs §4.4	Exclusion reasons differ: two classes in the figure vs three classes in the text	Align the figure box text with the three-class rule
11	§4.2–4.3 and Figure 1	Probe-added records (approx. 2,000 vs 1,287) not reconciled through screening; the 469-record set is undefined	Add probe strata to the flow; define the 469-record set
12	Table 3 vs §5.4	“Poisoning 22” vs “ai_poisoning 19”: an unexplained three-record difference	Footnote the construct difference (threat-term mention vs AI-threat engagement coding)
13	§4.3 vs §5.4	Wilson intervals on a non-random sample; no cutoff sensitivity	Drop or relabel the intervals; add a top-150/200 sensitivity analysis
14	Data availability (p. 24)	No URL or DOI for the reproducibility package	Add a resolvable repository link
15	CRediT statement (p. 24)	“(To be finalised.)” left in the text	Finalise the contribution statement

7. Recommendation and Revision Checklist

I recommend Major Revision. The core asymmetry finding is directionally credible: it survives the authors’ own adjudication, false-negative audit and subset sensitivities, and the lifecycle and release-rate findings are plausible and valuable. But the quantitative framing (a non-random sample presented with sampling intervals), the reliability of the headline fields ($\kappa = 0.41$ and 0.14), the definitional dependence of the sharpest claims, and the incomplete referencing and flow reporting are all substantive rather than cosmetic. None of the required work threatens the central result; most of it strengthens the paper’s own thesis. Table 3 gives my verdict on each headline claim and the change it needs.

Table 3: Verdict on the five headline claims

Claim	Verdict	Required change
AI is defensive in 70/83; attack-side engagement is a minority	Supported in direction	Keep; add an “of the retained corpus” qualifier at every headline use
35/83 (42.2%, Wilson CI 32–53%) engage any AI-enabled threat	Conditionally supported	Drop the CI or justify a superpopulation model; add a top-150/200 cutoff sensitivity analysis
11/83 engage one threat substantively	Fragile (κ = 0.14; one agreed cell)	Label as exploratory; remove from the Highlights; enlarge the dual-coded sample
The twin is an explicit attack target in exactly one study	Definition-dependent	Rephrase as “at most one study under a strict codebook definition”; add a loose-definition sensitivity coding
Coverage is 66/83 operation, 1 design, 0 commissioning	Low-reliability field (κ = 0.32; coder-1 default bias)	Re-analyse the multi-label records; attach the caveat to agenda item P2

The following ten actions, in priority order, would in my judgement bring the manuscript to acceptance standard:

1. Complete the reference list: replace every “journal (2026)” venue placeholder with the actual venue, complete the author list of reference [7], and resolve the “Anonymous” authorship of reference [19].
2. Rework the abstract to about 250 words; fix the misplaced parenthetical, the “moderate to substantial” self-contradiction and the κ-collapse wording; remove “(fragile, original)” from the Highlights.
3. Resolve the probe-yield accounting: add probe strata to the PRISMA flow, report probe-only membership in the retained 100, state the search execution date, and define the 469-record set.
4. Strengthen or bound the reliability evidence: enlarge the blind dual-coded sample for ai_threat_overall, twin_role and lifecycle_stage; report κ with bootstrap intervals; assign headline-field adjudication to coder 2 or a third party.
5. Add a cutoff sensitivity analysis (top 150 and top 200) for the retained-100 rule, and drop or re-derive the Wilson intervals as discussed in W1.
6. Run a loose-definition sensitivity coding for twin_role (W3) and rephrase “exactly one study” accordingly in the abstract and highlights.

7. Add a denominators table to Section 5 and reconcile the poisoning 22-versus-19 discrepancy.
8. Apply a lightweight quality or risk-of-bias appraisal, or elevate the non-appraisal disclosure to the abstract-level limitations.
9. Run a Fisher exact test (or report an interval) for the review-share comparison in §5.4, replacing the current “unsupported by any test” statement.
10. Add a resolvable URL or DOI for the reproducibility package in the Data availability statement, and attach the package to the submission so the coding sheets and the reliability script can be audited.

I close on the manuscript’s strongest feature. The culture of disclosure in this paper — unfavourable κ values, flagged post-hoc analyses, declared adjudication precedence, a bounded audit — is what reporting practice in this field should look like, and it makes revision unusually collaborative: the authors have already done the hard epistemic work of identifying their own weak points. What remains is to align the headline rhetoric with that honesty, complete the reporting apparatus, and deepen the reliability evidence for the fields the paper most depends on. With those changes, this review would be a genuinely useful benchmark for how systematic reviews of AI-security literatures should be reported.